

Probabilistic Separation Budgets for Primary Paths

Separation failure as a receipt-grade (ε, δ) adjunct (Series Synthesis)

Unified archive note v0.06 (2026-02-28; archive v130)

Abstract

Many reader-privacy deployments rely on a *separation* primary path: one role observes client-side metadata while a different role observes destination-side metadata, and the claim is interpreted only if these roles do not collude within the threat window. Synthesis 22 standardizes a digest-bound separation manifest so this assumption is explicit and diffable.

This note adds a minimal quantitative adjunct: if a deployment is willing to publish an upper bound ρ on separation failure (collusion / handle-join) within the declared threat window, then a “0-bit” separation intent yields a guarded *additive* bound

$$\Pr[X \in E \mid K = k] \leq \Pr[X \in E] + \rho \quad \text{for all events } E \text{ and all } k.$$

Equivalently, $\text{TV}(\text{Law}(X \mid K = k), \text{Law}(X)) \leq \rho$ for all k . This can be carried as a receipt-grade checkpoint $(0, \rho)$ in the series’ existing (ε, δ) calculus *without changing the public receipt schema*.

Scope. This is an interface note. It does *not* provide a universal way to estimate collusion risk. It only specifies how a published *upper bound* on separation failure can be carried as a digest-bound adjunct and how it propagates to endpoint interpretation.

Imports. Separation manifests are owned by Synthesis 22.[1] Receipt line-item semantics are owned by Synthesis 12 and equivalence/drift semantics by Synthesis 18.[2, 3] Threat windows are owned by Synthesis 4.[4] Endpoint meaning is owned by Anonymity A and the endpoint bridge.[6, 7] Horizon composition and checkpoint accounting are centralized in Synthesis 19.[11] A concrete instantiation of a separation-guard primary path appears in the worked example (Synthesis 17).[8]

Exports. A minimal “probabilistic separation” adjunct: a digest-bound ρ field bound to a separation manifest and threat window, plus an editorial rule for how to interpret it as a δ term.

1 The model

Fix a guarded exposure surface (ENF-ID) and threat window (TW-ID).[5, 4] Under the separation manifest, the deployment intends that the primary-path observation X is *key independent* on that surface (a “0-bit” linkability surface). In practice, the separation can fail (collusion, shared handles, correlated telemetry joins).

Definition 1 (Separation failure indicator). *Let $F \in \{0, 1\}$ be a (possibly hidden) indicator for whether separation is violated during the threat window for the guarded surface. A deployment publishes an upper bound*

$$\Pr[F = 1] \leq \rho$$

for some $\rho \in [0, 1]$. The bound is interpreted as being conditioned on the same public state scope declared by `state_decl_id` when applicable.[3]

Remark 1 (Why ρ is receipt-grade). *The series already interprets one-vs-mixture privacy via (ε, δ) budgets and composes them via checkpoint calculus.[6] Publishing ρ therefore does not introduce a new metric family; it introduces a failure-probability adjunct that deterministically maps to the checkpoint $(0, \rho)$ under the same endpoint semantics.*

2 From probabilistic separation to an (ε, δ) bound

The key observation is: if separation holds, the guarded surface is key independent; if it fails, we assume nothing. This yields an immediate additive bound whose parameters are determined solely by ρ .

Lemma 1 (ρ -separation yields a guarded $(0, \rho)$ checkpoint). *Let K be the hidden key (or destination class) and let X be the declared primary-path observation on the guarded exposure surface. Let $F \in \{0, 1\}$ indicate separation failure during the threat window. Assume: (i) conditioned on $F = 0$, the law of X is independent of K (the “0-bit” separation intent), and (ii) for all k , $\Pr[F = 1 \mid K = k] \leq \rho$. Then for any prior on K and any event E ,*

$$\Pr[X \in E \mid K = k] \leq \Pr[X \in E] + \rho \quad \text{for all } k,$$

and therefore $\text{TV}(\text{Law}(X \mid K = k), \text{Law}(X)) \leq \rho$ for all k .

Proof. Fix an event E and define $a \stackrel{\text{def}}{=} \Pr[X \in E \mid F = 0]$, $\rho_k \stackrel{\text{def}}{=} \Pr[F = 1 \mid K = k]$, and $b_k \stackrel{\text{def}}{=} \Pr[X \in E \mid K = k, F = 1]$. Key-independence under $F = 0$ makes a independent of k . Then

$$\Pr[X \in E \mid K = k] = (1 - \rho_k)a + \rho_k b_k = a + \rho_k(b_k - a).$$

Let $t_k \stackrel{\text{def}}{=} \rho_k(b_k - a)$. Since $b_k \in [0, 1]$ and $\rho_k \leq \rho$, we have $t_k \in [-\rho a, \rho(1 - a)]$. Let $\bar{t} \stackrel{\text{def}}{=} \mathbb{E}[t_K]$ under the prior on K . Then $\Pr[X \in E] = a + \bar{t}$, so

$$\Pr[X \in E \mid K = k] - \Pr[X \in E] = t_k - \bar{t} \leq \max t - \min t \leq \rho.$$

□

Remark 2 (On multiplicative displays). *Lemma 1 is already an (ε, δ) -style statement with $\varepsilon = 0$ and $\delta = \rho$. If a consumer insists on writing a multiplicative factor, note that $(0, \rho)$ trivially implies (ε, ρ) for any $\varepsilon \geq 0$. There is no canonically “best” ε without additional assumptions.*

Remark 3 (Endpoint meaning). *Lemma 1 is a guarded statement: with probability at most ρ , the separation claim may fail completely. Downstream endpoint interpretation should therefore treat $(0, \rho)$ as an additive checkpoint in the same style as approximate DP.[6] Publishing a larger bound $\rho' \geq \rho$ is a monotone relaxation (weaker guarantee); publishing a smaller bound is a tightening that should be backed by evidence/replay.*

3 How to publish ρ without changing the schema

Synthesis 12 deliberately keeps the receipt schema minimal.[2] Accordingly, the probabilistic separation adjunct is carried as a digest-bound field adjacent to the receipt:

- The primary-path line item includes the `sep_manifest_id` (or equivalent manifest digest pointer) as described in Synthesis 22.[1]
- The same line item carries `rho_upper` inside `obs_model` (or inside `knobs`) and tags whether it is a *policy bound* or an *evidence-backed estimate*.
- Drift rule (aligned to Synthesis 18): treat any *tightening* of `rho_upper` as replay/evidence-required; treat a *relaxation* ($\rho' \geq \rho$) as a declared `lineage=coarsening` change that does not require new evidence but must still be published and diffed.[3]

Interpretation rule. Auditors and endpoint interpreters treat `rho_upper` as defining the guarded checkpoint $(0, \rho)$ via Lemma 1.

4 Research warranted (what this note does not solve)

This note assumes a published upper bound ρ . Estimating ρ in a way that is (i) meaningful, (ii) replayable, and (iii) resistant to adversarial “audit $\neq$ live” bias is an open engineering problem. A plausible publication shape is a conditional certificate (Synthesis 21) whose evidence object is an *evidence bundle* that makes separation failure mechanically checkable.[9, 10]

4.1 Toy bounds (when you are willing to assume a model)

This note is deliberately not a “collusion-risk estimator,” but it is still useful to record a few *model-to- ρ* translations that are short enough to review. These are only meaningful if the corresponding assignment/rotation policy is committed in the separation manifest (Synthesis 22).[1]

Lemma 2 (Independent compromise model $\Rightarrow$ multiplicative ρ bound). *Assume the separation manifest pins a policy in which a lookup chooses a relay instance R from a declared relay set and a router instance U from a declared router set, and the operator sets are declared disjoint. If the relay-side compromise event A_R and router-side compromise event A_U satisfy $\Pr[A_R] \leq p_R$, $\Pr[A_U] \leq p_U$, and A_R is independent of A_U , then the separation-failure probability (collusion / join of both sides) is bounded by*

$$\rho \leq \Pr[A_R \wedge A_U] \leq p_R p_U.$$

Remark 4 (Finite-adversary clause as a special case). *If the manifest asserts “at most t_R of m_R relay operators are compromised” and the relay is selected uniformly at random from that set, then one may take $p_R = t_R/m_R$ (and similarly $p_U = t_U/m_U$) in the lemma above. This is not a theorem about real adversaries; it is simply the translation from a committed assignment policy and a committed adversary-count clause into an upper bound.*

Remark 5 (Threat-window aggregation). *If the threat window covers d independent (or union-bounded) epochs with per-epoch separation failure bounds $\rho_1, \dots, \rho_d$, then $\Pr[\exists i : F_i = 1] \leq \sum_i \rho_i$. In particular, under a uniform per-epoch bound ρ_{epoch} one may publish the conservative window bound $\rho \leq \min\{1, d \rho_{\text{epoch}}\}$. This is the correct place to encode “rotation helps” in a receipt-grade way: the manifest pins the rotation policy, and the published ρ is the resulting window-level bound.*

- **Handle-join and key-join detectors:** a logged set of forbidden joins (shared identifiers, shared TLS client certs, shared routing keys) over the threat window, with the detector definition digest-bound in the plan spec.
- **Administrative non-collusion attestations:** signed statements binding relay/router operators, billing accounts, and incident-response channels (useful but not sufficient alone).
- **Cryptographic separation anchors:** evidence that relay- and router-side keys are rooted in disjoint HSM domains or disjoint issuing CAs for the window, so “shared handle” joins are detectable.
- **Relay/router log cross-checks:** consistency checks that the sets of requests/epochs seen by each role match the declared interface without revealing per-request linkage (e.g., via digest commitments to per-epoch aggregates).
- **Audit $\neq$ live hardening:** replay plans that sample both audit and live traffic to detect conditional failures (e.g., collusion only when audit flags are absent).

References

- [1] Anonymous. Primary-Path Separation Manifests: Guard Objects for Non-Collusion Assumptions in Reader-Privacy Receipts. Synthesis 22, The-One-True-Archive (2026).
- [2] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Synthesis 12, The-One-True-Archive (2026).
- [3] Anonymous. Digest-Bound Replay Plans and State Declarations: Equivalence, Minimality, and Change Control. Synthesis 18, The-One-True-Archive (2026).
- [4] Anonymous. Threat Models and Repetition Windows for Anonymous DHTs. Synthesis 4, The-One-True-Archive (2026).
- [5] Anonymous. Trace Interfaces for Anonymous DHTs. Synthesis 3, The-One-True-Archive (2026).
- [6] Anonymous. Odds Inflation, Privacy Loss, and Endpoint Anonymity Bounds. Anonymity series Paper A, The-One-True-Archive (2026).
- [7] Anonymous. Endpoint Metrics Bridge: Odds Inflation, PML/MaxL, and Identification Sizing. Synthesis 5, The-One-True-Archive (2026).
- [8] Anonymous. Worked Example: Instantiating a Tiered Reader-Privacy Receipt. Synthesis 17, The-One-True-Archive (2026).
- [9] Anonymous. Conditional Per-Step Budget Certificates: A Receipt-Grade Interface for Adaptive Horizon Claims. Synthesis 21, The-One-True-Archive (2026).
- [10] Anonymous. Auditor Replay for Anonymous-DHT Receipt Line Items: A Mechanical Checklist for Digest-Bound Claims. Synthesis 20, The-One-True-Archive (2026).
- [11] Anonymous. Receipt Accounting Rulebook: Checkpoints, Composition, and Ledger-Friendly Summaries. Series synthesis note (Synthesis 19), 2026. (This archive.)